

Optically Integrated 1.12-Tb/s/λ DSP-free Communication and Vibration Sensing over 10-km Hollow-core Fiber for Scale-across Datacenter Interconnect

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Abstract We demonstrate the integration of communication and forward-vibration-sensing in a DSP-free coherent receiver for low-cost, high-reliability scale-across networks. Using 10-km hollow-core fiber and pilot-tone-assisted optical phase-locked loop, we achieve simultaneous proactive sensing and communication with a record distance–rate product of 11.2-Tb/s-km/λ, without standard receiver DSP.

Introduction

The rapid growth of large-scale AI distributed training places increasing demands on scale-across datacenter interconnect (DCI) to support extremely high throughput, low power consumption, low latency, and high reliability. From a throughput perspective, coherent communication systems are promising candidates, while the digital signal processing (DSP) required in conventional coherent receivers is often criticized for its cost and power consumption [1]. DSP-free coherent receivers therefore represent a promising approach, enabling coherent communication to be extended to scale-across DCI scenarios [2–5]. It has been well established that hollow-core fiber (HCF) is well suited for DCI owing to its low loss and latency [6]. As less reported, HCF is also particularly attractive for DSP-free links due to its ultra-low chromatic dispersion (CD), which enables the use of readily available C-band coherent components.

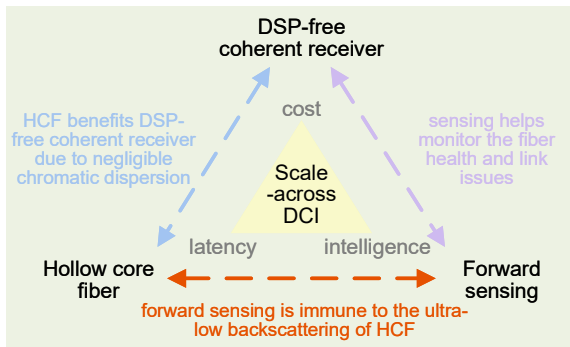


Fig. 1: Schematic of ISAC in DSP-free scale-across DCI.

From a reliability perspective, the stability and serviceability of DCI links remain unsolved challenges, with a particular focus on vibrations induced by fiber breaks and damages [7]. In the context of digitally twinning the physical state, distributed fiber-optic sensing (DFOS) offers an effective solution for ensuring stable link operation and monitoring physical integrity [8]. There are two main limitations to embed DFOS function into a HCF-based DSP-free DCI link: 1) It is difficult to integrate Rayleigh-based sensing on HCF due to its ultra-low scattering coefficient [9]. Hence, forward sensing becomes a feasible way to enable sensing in HCF-based link [10]. 2) The existing

forward sensing methods based on coherent receiver either rely on digital equalizers [11] or carrier recovery [12], which is not available in DSP-free receiver. However, as the key to remove phase noise (PN), phase-locked loop (PLL) has the potential to simultaneously provide the sensing information.

To address these issues, we present a seamlessly integrated sensing and communication (ISAC) method for DSP-free DCI as illustrated by Fig. 1. We demonstrate simultaneous C-band DSP-free communication with record distance-rate product of 11.2 Tb/s-km/λ and proactive sensing based on optical PLL (OPLL) and HCF.

Designed pilot-tone-assisted OPLL and its enabled forward vibration sensing

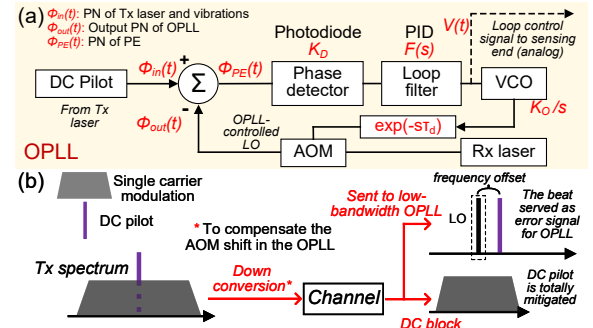


Fig. 2: (a) The OPLL mathematic model and its sensing principle. (b) Pilot-assisted OPLL method.

Fig. 2a illustrates the structure of OPLL and the forward vibration sensing principle. Suppose the input PN $\phi_{in}(t)$ is composed of laser noise with power spectral density (PSD) $S_{\phi}(f) = S_{\nu}(f)/f^2$ and vibration noise $\phi_{vib}(t)$, the phase error (PE) is the beat with the output of OPLL after passing the PLL modeling,

$$\phi_{PE}(s) = \phi_{in}(s) - \phi_{out}(s) = \phi_{in}(s)/(1 + L(s)e^{-s\tau_d}) + n(s), \quad L(s) = K_D(K_P + K_I/s)K_O/s, \quad (1)$$

where $L(s)$, τ_d , K_P , K_I , $n(s)$ denote the open-loop gain, loop delay, proportional gain, integral gain, and noise, assuming a second-order PLL with $F(s) = K_P + K_I/s$. It is found that the loop control signal of PID is proportional to ϕ_{PE} ,

$$V(t) = \alpha \cdot \mathcal{L}^{-1}\{K_D(K_P + K_I/s)\phi_{PE}(s)\}, \quad (2)$$

α and $\mathcal{L}$ are the proportion coefficient and Laplace

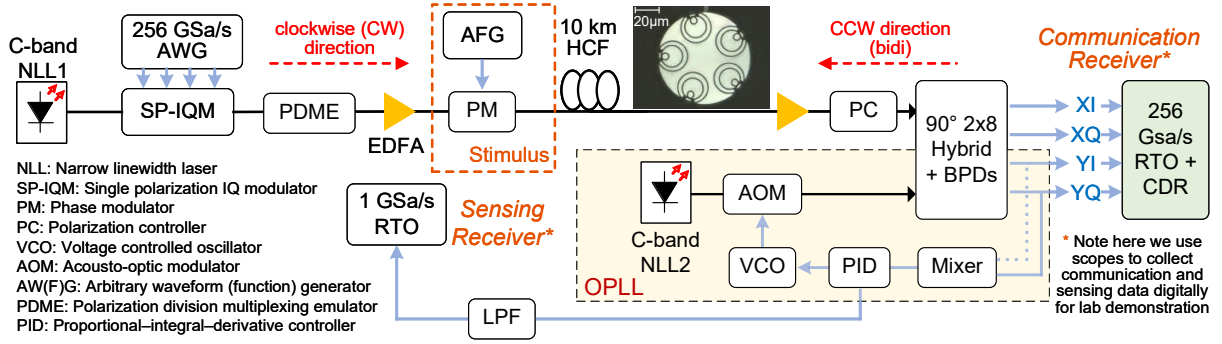


Fig. 3: Experimental setup. Note that only CW direction of bi-directional (Bidi) setup is plotted.

transform, respectively. $V(t)$ can indicate the fluctuation of $\phi_{in}(t)$ and therefore $\phi_{vib}(t)$ through Eq. 1. Note that we are not using the digitally demodulated phase profile. Instead, we exploit the intrinsic analog loop control signal of an OPLL for forward sensing without any modifications to the communication receiver. It should be pointed out that, since $V(t)$ is filtered vibration information, the proposed sensing method cannot fully reconstruct the vibration waveform, unlike traditional distributed acoustic sensing (DAS). However, for scale-across links, the proposed method is capable of proactively detecting fiber-damage-induced vibrations, similar to state-of-polarization (SOP)-based fiber-break sensing demonstrated in [7].

Fig. 2b details the design of our pilot-tone-assisted OPLL. This design incorporates several enhancements over standard implementations: 1) By placing the pilot at DC (zero frequency) instead of using a conventional low-frequency tone, we can use a simple DC-block for removal. This alleviates the trade-off between pilot power and overall communication performance. 2) We utilize electrical signal splitting for the feedback loop instead of optical splitting. This approach is more effective at suppressing slow PN induced by environment. 3) To be compatible with single-carrier coherent modulation (which can be decomposed to standard PAM signals after I/Q separation), we chose not to use dual-subcarrier modulation (DSM), despite DSM could improve pilot-tone purity.

Experimental setup

Fig. 3 shows our experimental setup to verify method. An AWG produces the QAM signal with a pilot tone, which is then applied to a SP-IQM followed by a PDME to form a dual-polarization signal. After transmission through 10-km support-tube HCF (ST-HCF), the signal is detected by the DSP-free receiver. As this work focuses on the OPLL, the polarization is manually adjusted using a PC to separate the polarization states. In practical implementations, this process can be automated using an automatic polarization controller [13]. After fine-tuning the PID servo gain and switching the OPLL from the open-loop to the closed-loop state, the LO dynamically adjusts its phase via the AOM (40 MHz bandwidth, 200 MHz shift). The four datastreams are then captured by a real-time oscilloscope (RTO, Keysight UXR1102A). Here, a digital clock-data recovery (CDR) algorithm is employed to emulate the operation of a

physical CDR module. For the sensing side, we use the phase modulator (PM, EOSPACE) of 5 V V_{pi} to simulate the vibration source, and the relationship between strain/m and corresponding phase variation can be found in [14]. A 500-kHz low-pass filter (LPF) is used to filter out noise outside the desired frequency range, as most practical vibration frequencies are below 500 kHz.

Results and discussions

We first assess the effectiveness of pilot-tone-assisted OPLL. As depicted in Fig. 4a, we tested two types of NLLs (one with <100 Hz intrinsic linewidth [15] and the other with <1 kHz [16]) and one commercial integrable tunable laser assembly (ITLA) for use in OPLL. The long-term frequency offset tests revealed that OPLL can be easily locked with NLLs, while ITLA fails to achieve locking. As is well known, DFOS typically requires sub-kHz-linewidth lasers to distinguish vibration-induced PN from laser-induced PN. The results indicate that OPLL-based systems likewise require stabilized and narrow-linewidth lasers, therefore lasers for fiber sensing are more than sufficient for use in OPLL. Furthermore, by utilizing a frequency-synchronous network from a single seed laser in DCI network[3], forward sensing and OPLL can share stabilized lasers at a reduced cost.

A 1 Vpp burst 100-kHz sine stimulus with a 10 μ s duration and a 1 kHz repetition rate is initially applied to the PM to simulate vibrations. As depicted in Fig. 4c, the vibration-induced burst-like signal is clearly visible above the noise floor. Locating vibrations requires bi-directional transmission and correlation. Two cases of bi-directional tests are analyzed: one using a single pilot tone and the other using communication data. The results show that both cases yield the same correlation peak, suggesting that data has negligible effects on the sensing performance. Since the OPLL has a narrow bandwidth and high-speed data has a wide bandwidth, the OPLL employing a single pilot tone is robust to communication signals. The correlation of both sides is shown in the inset of 4c, indicating a time delay of 33.2 μ s, which corresponds to the traveling time over the ST-HCF.

We further modify the stimulus to simulate various vibrations, including intensity, duration, and waveforms. The results are depicted in Fig. 4d-f. Additionally, the PSD of the PE with different waveforms is shown in Fig. 4g. The PE is measured and analyzed by capturing the I and Q data in the

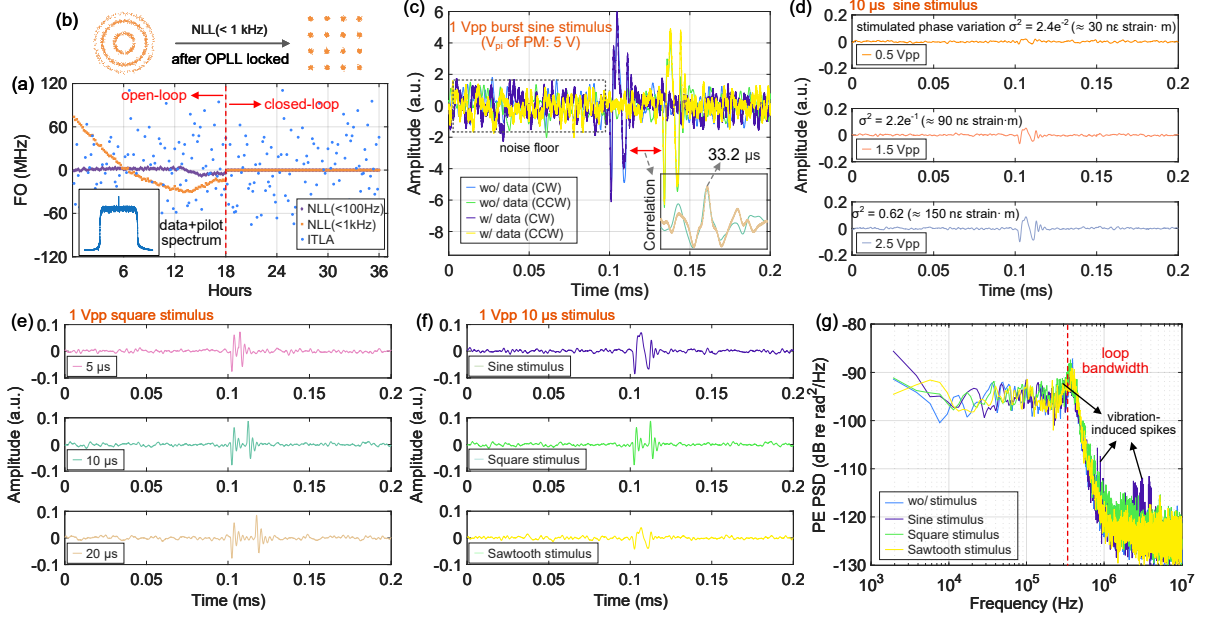


Fig. 4: (a) Long-term frequency offset (FO) test. (b) 16-QAM constellation before/after locked. (c) Bi-directional profiles of $V(t)$ with data on and off. (d)-(f) Sensing patterns of $V(t)$ with different stimulus. (g) PSD of PE profile with stimulus of different waveforms.

RTO. The results indicate that, in the presence of stimulus, the PSD of PE exhibits multiple spike-like spectral components, which correspond to the applied vibration waveforms.

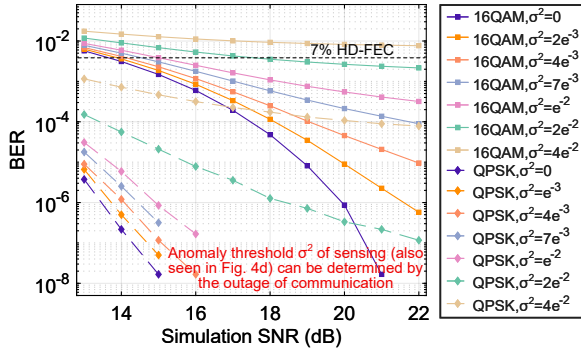


Fig. 5: Simulation of instantaneous BER degradation under various vibration-induced phase variance σ^2 .

In practice, proactive sensing can be triggered when the vibration-induced phase variance σ^2 exceeds a predefined anomaly threshold (AT). In the proposed ISAC system, AT may be determined based on communication performance degradation. For instance, when the bit error rate (BER) exceeds a specified limit, proactive sensing can be activated to identify fiber-damages-induced vibrations. Accordingly, in Fig. 5, we simulate the instantaneous communication performance degradation of 16-QAM and QPSK under various values of σ^2 , while sweeping the SNR in AWGN system model. The results indicate that AT can be set differently under cases. In systems with lower SNR, vibration-induced phase variations generally have a reduced impact on BER.

We then evaluate the 16-QAM performance of the DSP-free coherent receiver. The demodulated data obtained using OPLL are compared with those processed using digital carrier phase recovery (DCPR) under identical conditions, as

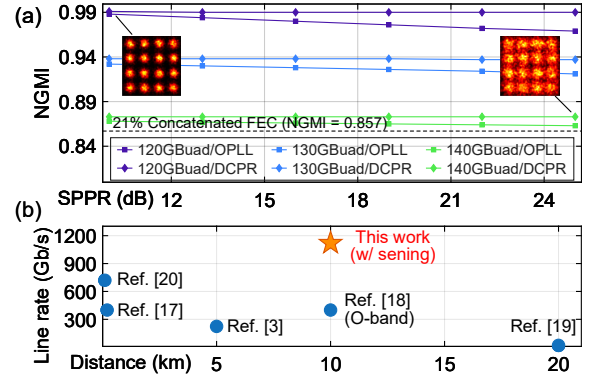


Fig. 6: (a) Experimental results of telecom performance under different SPPRs. (b) Distance-rate product comparisons.

shown in Fig. 6a. The results indicate that the OPLL achieves performance comparable to that of DCPR. By sweeping the signal-to-pilot power ratio (SPPR), we observe that the normalized generalized mutual information (NGMI) slightly degrades as the SPPR increases. This behavior is attributed to the reduced SNR of the pilot tone used for OPLL. Since the 140-Gbaud case still exceeds the 21% concatenated forward error correction (FEC) threshold, a net-rate of 925 Gb/s is achieved. Furthermore, the transmission distance and line rate of recent DSP-free communication experiments are compared in Fig. 6b. Thanks to the ultra-low accumulated CD value of ST-HCF, the distance-rate product is higher than that of reported systems [3, 17–20].

Conclusion

We leverage ST-HCF and OPLL to simultaneously enable proactive vibration sensing and DSP-free communication, achieving a record distance-rate product of 11.2 Tb/s-km/ λ . The proposed ISAC method is promising to provide high throughput, low power consumption, low latency, and high reliability in large-scale AI scale-across networks.

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